ML LAB EXPT – 4

Aim of the Experiment:

For a given set of training data examples stored in a .CSV file, implement and demonstrate the Find-S algorithm to output a description of the set of all hypotheses consistent with the training examples.

```python
import pandas as pd

def find_s_algorithm(data):
    # Initialize the hypothesis with the first positive instance
    hypothesis = data.iloc[0, :-1].values.tolist()

    # Iterate through the remaining instances
    for index, row in data.iterrows():
        if row[-1] == 'Yes':  # Positive instance
            for i in range(len(hypothesis)):
                if hypothesis[i] != row[i]:
                    hypothesis[i] = '?'
    return hypothesis

# Load the training data from a CSV file
try:
    training_data = pd.read_csv('training_data.csv')
except FileNotFoundError:
    print("Error: training_data.csv not found.")
    exit()


# Run the Find-S algorithm
result_hypothesis = find_s_algorithm(training_data)
print("The most specific hypothesis is:", result_hypothesis)
```

Output:

The most specific hypothesis is: ['>9', 'Yes', '?', 'Good', '?', '?']